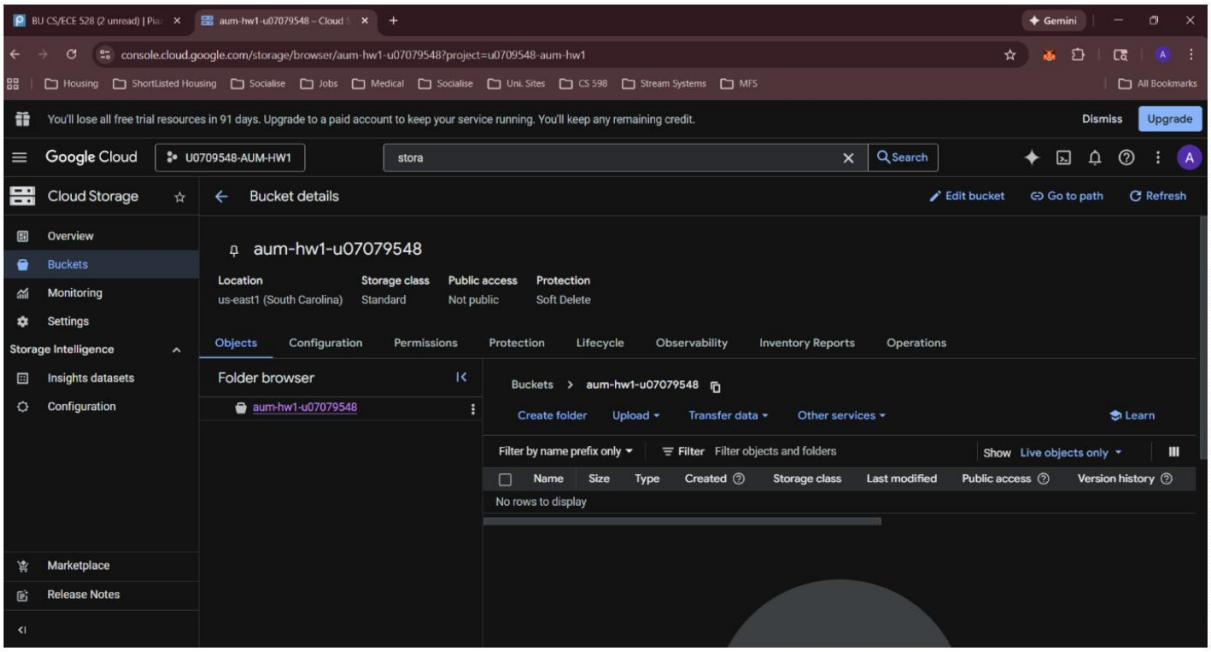
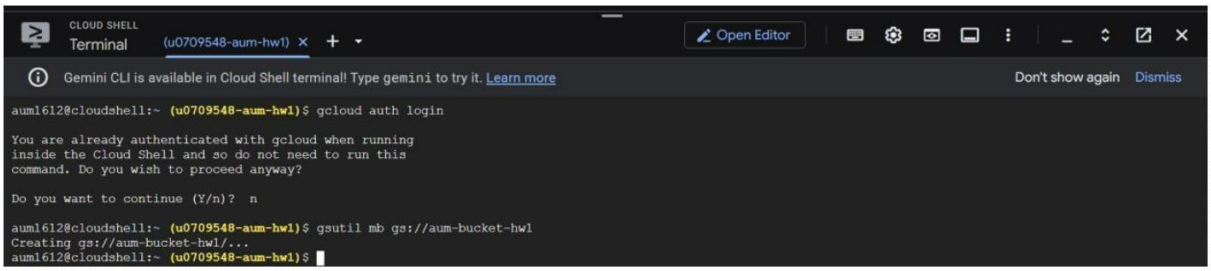
AUM GHELANI – U07079548 – HW1
CLOUD COMPUTING

1) Make a storage bucket



2) Make a directory inside your bucket



3) List the directory inside your bucket

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gsutil cp /dev/null gs://aum-bucket-hw1/directoryforhw1/
Copying file:///dev/null [Content-Type=application/octet-stream]...
/ [1 files][    0.0 B/    0.0 B]
Operation completed over 1 objects.
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gsutil ls gs://aum-bucket-hw1/
gs://aum-bucket-hw1/directoryforhw1/
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

4) Copy a file into your directory

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ echo "CS528 HW1 test aum1612" > hw1.txt
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gsutil cp hw1.txt gs://aum-bucket-hw1/directoryforhw1/
Copying file://hw1.txt [Content-Type=text/plain]...
/ [1 files][   23.0 B/   23.0 B]
Operation completed over 1 objects/23.0 B.
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

5) List the file inside your directory

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gsutil ls gs://aum-bucket-hw1/directoryforhw1/
gs://aum-bucket-hw1/directoryforhw1/hw1.txt
gs://aum-bucket-hw1/directoryforhw1/null
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

6) Delete the directory (and the file inside it) in a single operation

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gsutil rm -r gs://aum-bucket-hw1/directoryforhw1/
Removing gs://aum-bucket-hw1/directoryforhw1/hw1.txt#1770161239977260...
Removing gs://aum-bucket-hw1/directoryforhw1/null#1770160858321816...
/ [2 objects]
Operation completed over 2 objects.
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

7) Make a VM

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud compute instances create hw1-vm --zone=us-central1-a
API [compute.googleapis.com] not enabled on project [u0709548-aum-hw1]. Would you like to enable and retry (this will take a few minutes)? (y/N)?  y

Enabling service [compute.googleapis.com] on project [u0709548-aum-hw1]...
Operation "operations/acf.p2-1097067927322-f8f29574-8cd9-4255-8bca-f0140c994da8" finished successfully.
ERROR: (gcloud.compute.instances.create) Retry
This may be due to network connectivity issues. Please check your network settings, and the status of the service you are trying to reach.
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud compute instances create hw1-vm --zone=us-central1-a
Created [https://www.googleapis.com/compute/v1/projects/u0709548-aum-hw1/zones/us-central1-a/instances/hw1-vm].
WARNING: Some requests generated warnings:
 - You are creating a global DNS VM. VM instances using global DNS are vulnerable to cross-regional outages. To reduce the risk of widespread service disrupti
on, use zonal DNS instead. Learn more at https://cloud.google.com/compute/docs/networking/zonal-dns

NAME: hw1-vm
ZONE: us-central1-a
MACHINE_TYPE: n1-standard-1
PREEMPTIBLE:
INTERNAL_IP: 10.128.0.2
EXTERNAL_IP: 35.225.229.192
STATUS: RUNNING
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

8) Suspend the VM

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud compute instances suspend hw1-vm --zone=us-central1-a
Suspending instance(s) hw1-vm...done.
Updated [https://compute.googleapis.com/compute/v1/projects/u0709548-aum-hw1/zones/us-central1-a/instances/hw1-vm].
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

9) Stop the VM

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud compute instances stop hw1-vm --zone=us-central1-a
Stopping instance(s) hw1-vm...done.
Updated [https://compute.googleapis.com/compute/v1/projects/u0709548-aum-hw1/zones/us-central1-a/instances/hw1-vm].
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

10) List all your VMs

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud compute instances list
NAME: hw1-vm
ZONE: us-central1-a
MACHINE_TYPE: n1-standard-1
PREEMPTIBLE:
INTERNAL_IP: 10.128.0.2
EXTERNAL_IP:
STATUS: TERMINATED
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

11) Delete the VM

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud compute instances delete hw1-vm --zone=us-central1-a
The following instances will be deleted. Any attached disks configured to be auto-deleted will be deleted unless they are attached to any other instances or
the `--keep-disks` flag is given and specifies them for keeping. Deleting a disk is irreversible and any data on the disk will be lost.
 - [hw1-vm] in [us-central1-a]

Do you want to continue (Y/n)?  Y

Deleted [https://www.googleapis.com/compute/v1/projects/u0709548-aum-hw1/zones/us-central1-a/instances/hw1-vm].
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

12) Make a service account and give it permissions to access your storage bucket

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud iam service-accounts create hw1-sa --display-name="HW1 aum1612 Service Account"
Created service account [hw1-sa].
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gsutil iam ch \
  serviceAccount:hw1-sa@u0709548-aum-hw1.iam.gserviceaccount.com:objectAdmin \
  gs://aum-bucket-hw1
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

13) List all your service accounts

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud iam service-accounts list
DISPLAY NAME: Compute Engine default service account
EMAIL: 1097067927322-compute@developer.gserviceaccount.com
DISABLED: False

DISPLAY NAME: HW1 aum1612 Service Account
EMAIL: hw1-sa@u0709548-aum-hw1.iam.gserviceaccount.com
DISABLED: False
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

14) Delete the service account

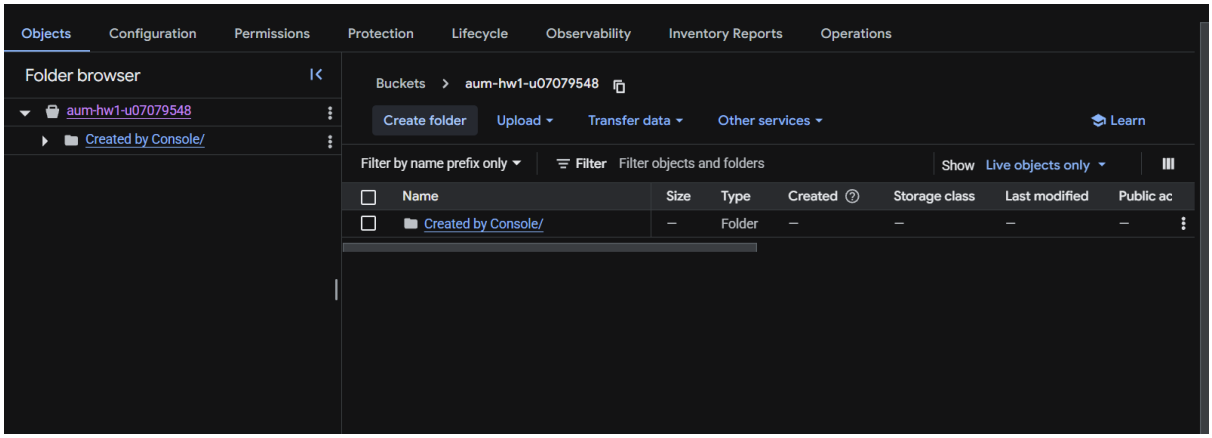
```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud iam service-accounts delete hw1-sa@u0709548-aum-hw1.iam.gserviceaccount.com
You are about to delete service account [hw1-sa@u0709548-aum-hw1.iam.gserviceaccount.com]

Do you want to continue (Y/n)?  Y

deleted service account [hw1-sa@u0709548-aum-hw1.iam.gserviceaccount.com]
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

15) Delete the storage bucket

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gsutil rm -r gs://aum-bucket-hw1
Removing gs://aum-bucket-hw1/...
aum1612@cloudshell:~ (u0709548-aum-hw1)$
```

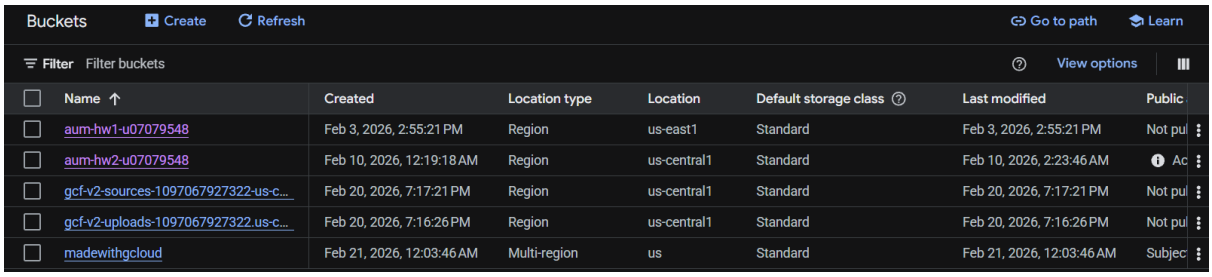
Question 1)

Console –



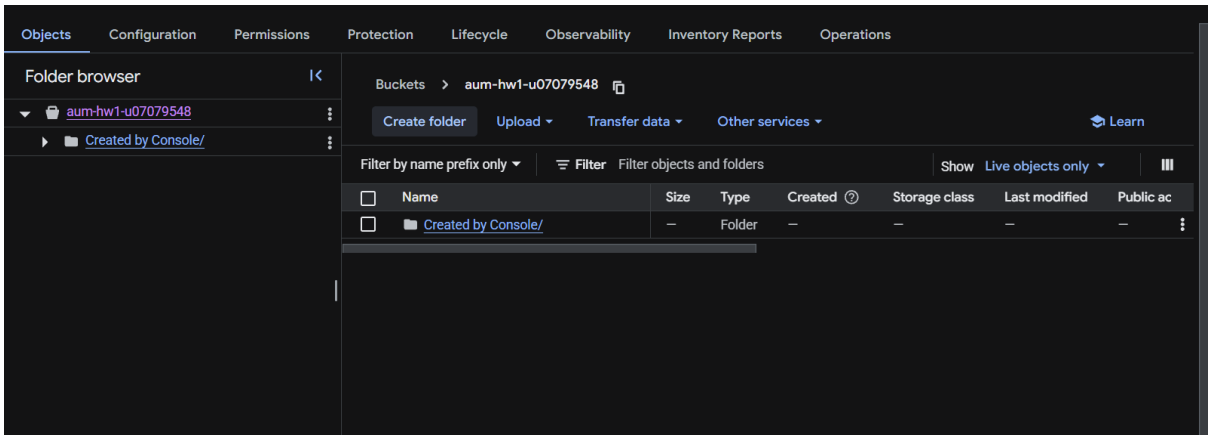
Gcloud –

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud storage buckets create gs://madewithgcloud
Creating gs://madewithgcloud/...
```



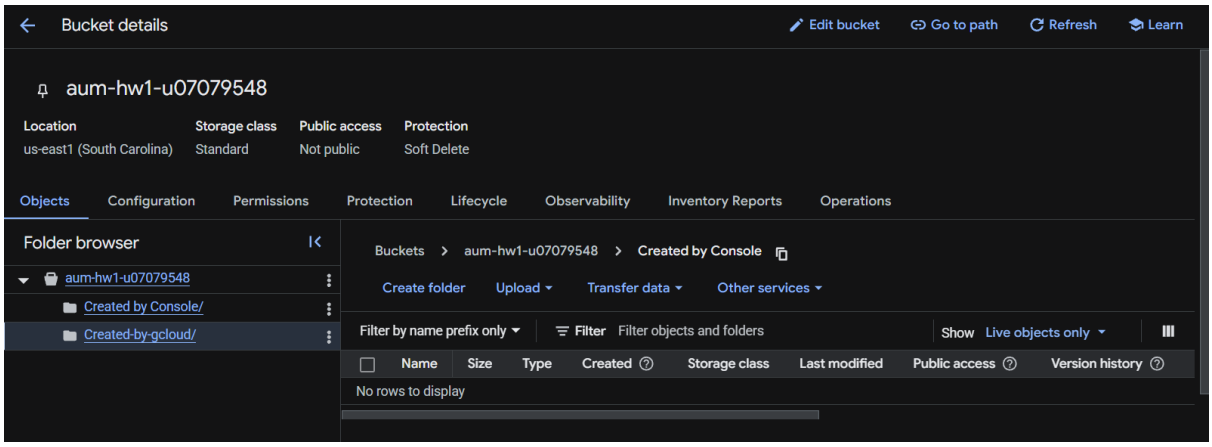
Question 2)

Console –



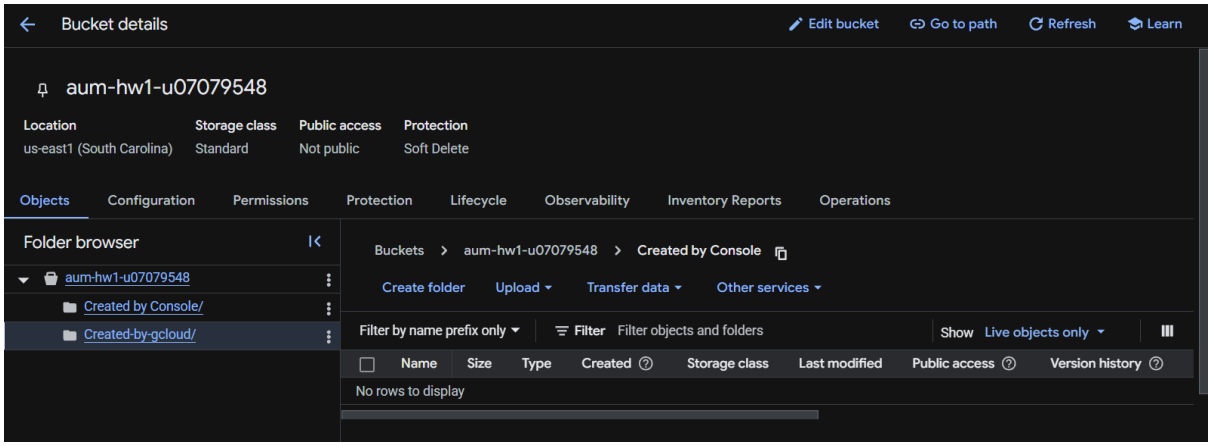
Gcloud –

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ touch gcloud.txt
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud storage cp gcloud.txt gs://aum-hw1-u07079548/Created-by-gcloud/
Copying file://gcloud.txt to gs://aum-hw1-u07079548/Created-by-gcloud/gcloud.txt
  Completed files 1/1 | 0B
```

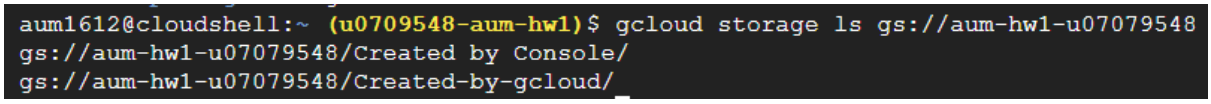


Question 3) List directories inside a folder

Console –

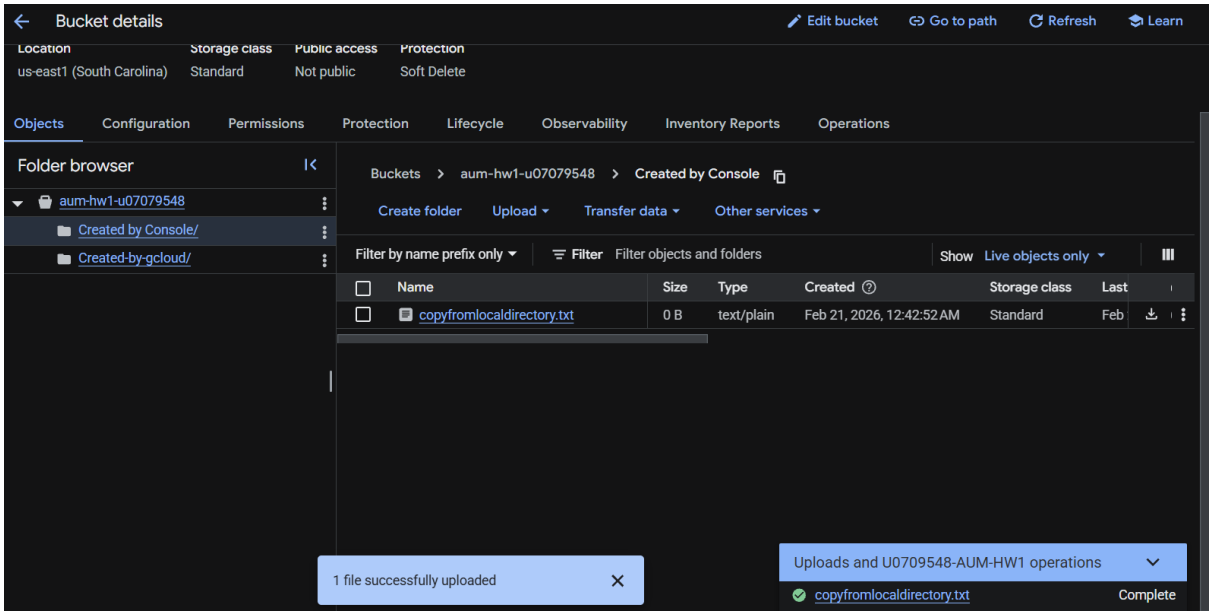


Gcloud –



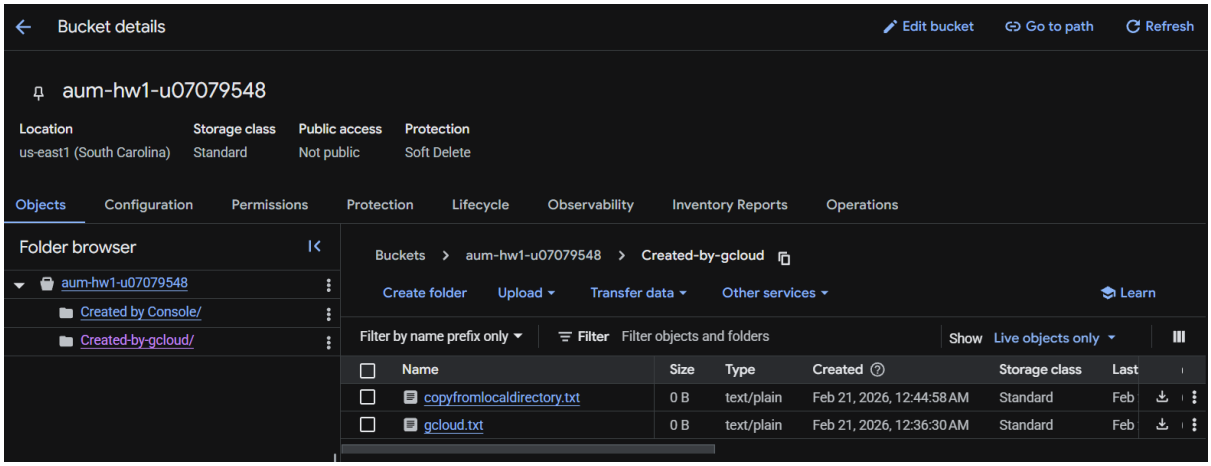
Question 4) Copy a file

Console –



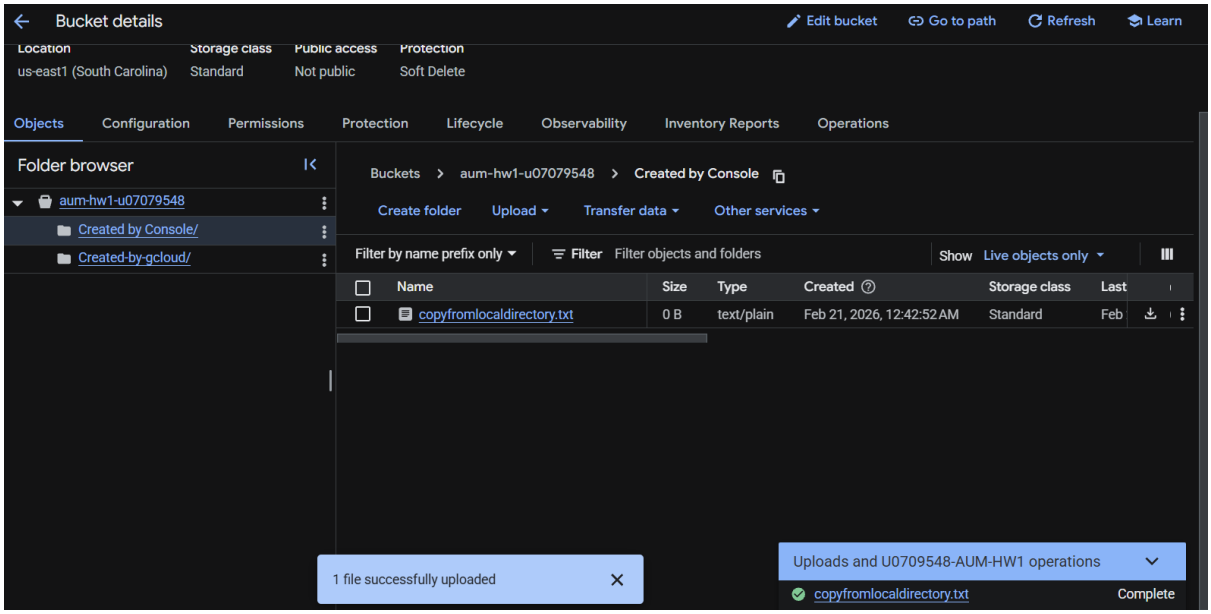
Gcloud –

```
PS D:\Masters Studies\CS 528 - Cloud Computing\HW\HW1> gcloud storage cp "D:\Masters Studies\CS 528 - Cloud Computing\HW
\HW1\copyfromlocaldirectory.txt" gs://aum-hw1-u07079548/Created-by-gcloud/
Copying file://D:\Masters Studies\CS 528 - Cloud Computing\HW\HW1\copyfromlocaldirectory.txt to gs://aum-hw1-u07079548/C
reated-by-gcloud/copyfromlocaldirectory.txt
  Completed files 1/1 | 0B


Updates are available for some Google Cloud CLI components.  To install them,
please run:
  $ gcloud components update



To take a quick anonymous survey, run:
  $ gcloud survey
```



Question 5) List files inside directory

Console –

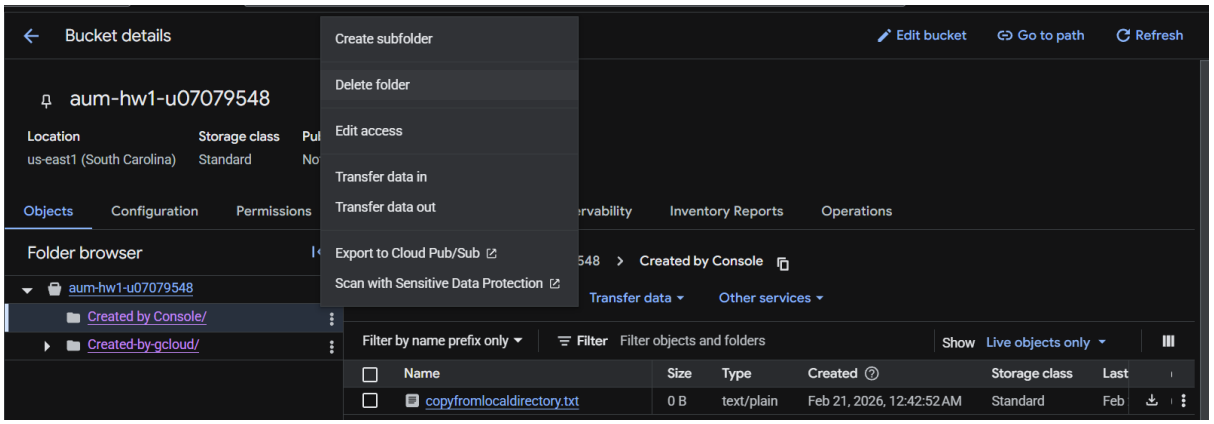


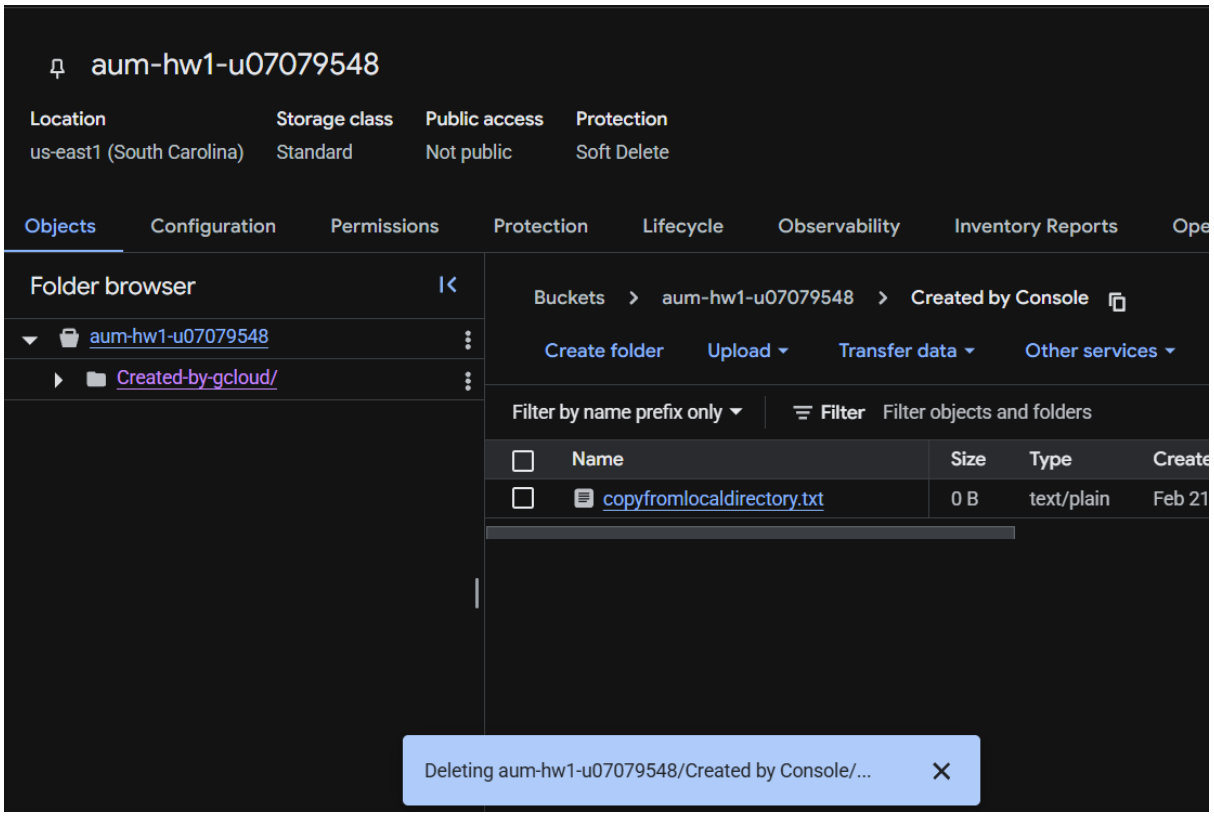
Gcloud –

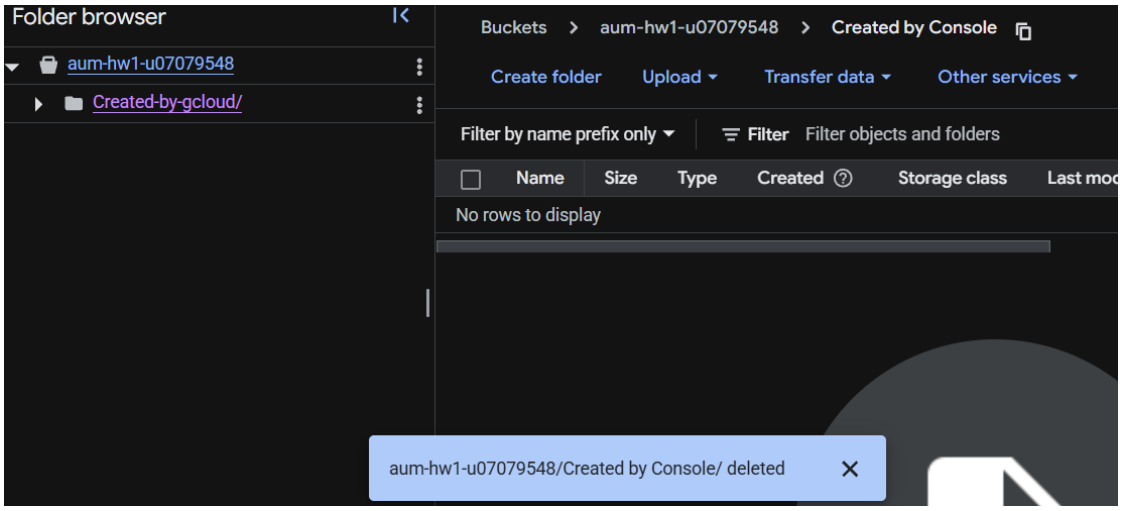
```
PS D:\Masters Studies\CS 528 - Cloud Computing\HW\HW1> gcloud storage ls gs://aum-hw1-u07079548/Created-by-gcloud/
gs://aum-hw1-u07079548/Created-by-gcloud/copyfromlocaldirectory.txt
gs://aum-hw1-u07079548/Created-by-gcloud/gcloud.txt
```

Question 6) Delete the directory and files inside single operation

Console –

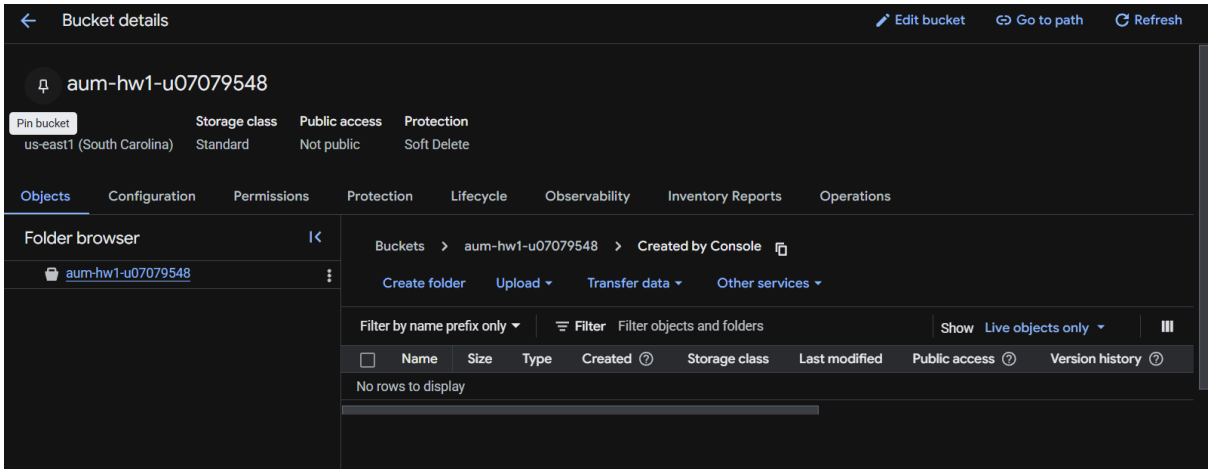






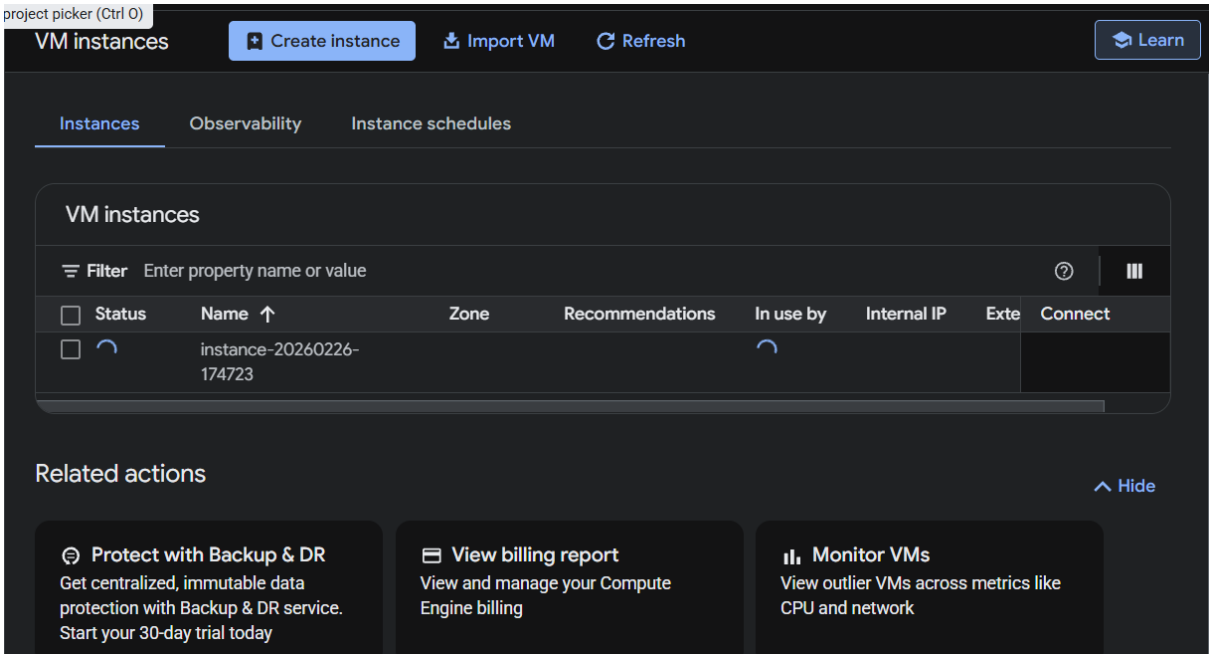
Gcloud –

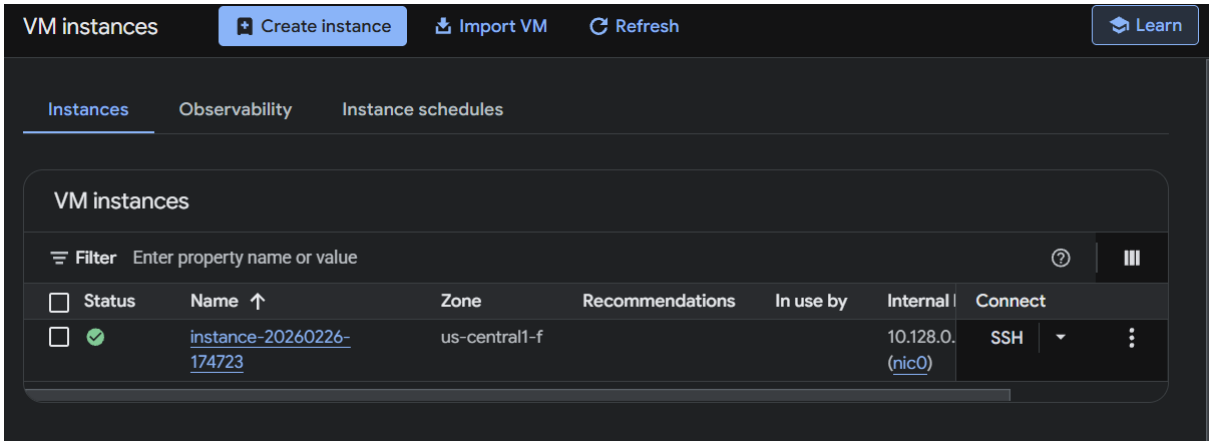
```
PS D:\Masters Studies\CS 528 - Cloud Computing\HW\HW1> gcloud storage rm -r gs://aum-hw1-u07079548/Created-by-gcloud/
Removing objects:
-Removing gs://aum-hw1-u07079548/Created-by-gcloud/copyfromlocaldirectory.txt#1772127898882398...
Removing gs://aum-hw1-u07079548/Created-by-gcloud/gcloud.txt#1772127390463847...
  Completed 2/2
```



Question 7) Create VMs

Console –



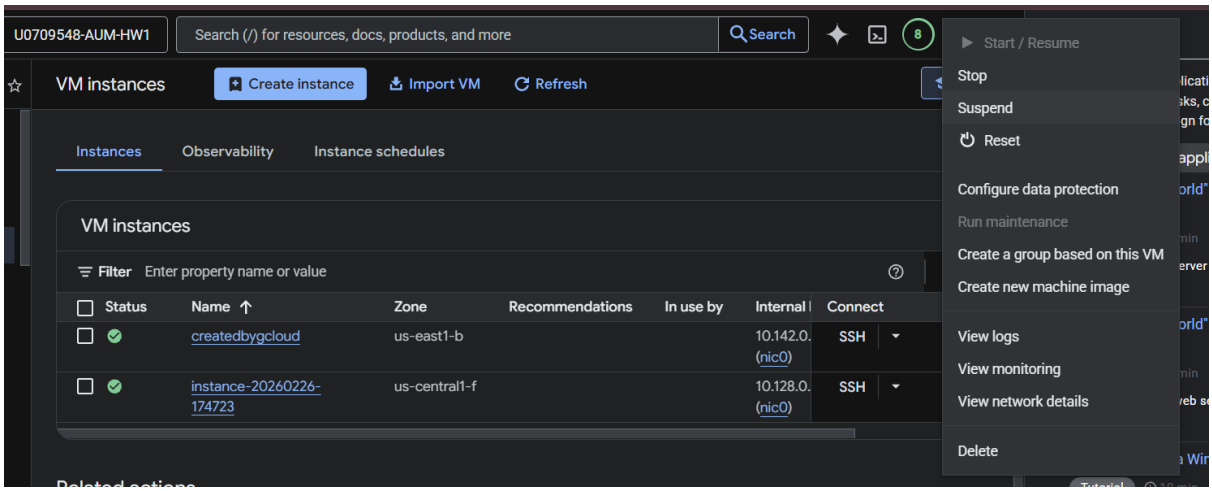


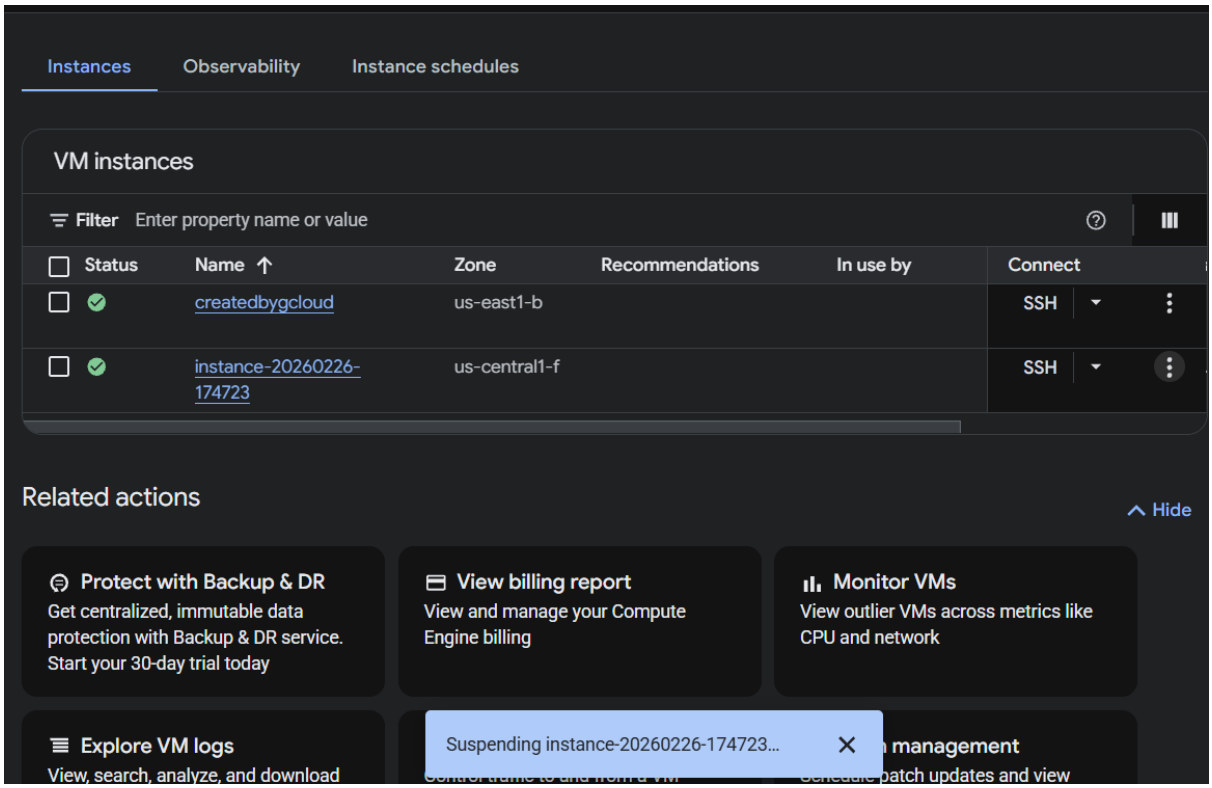
Gcloud –

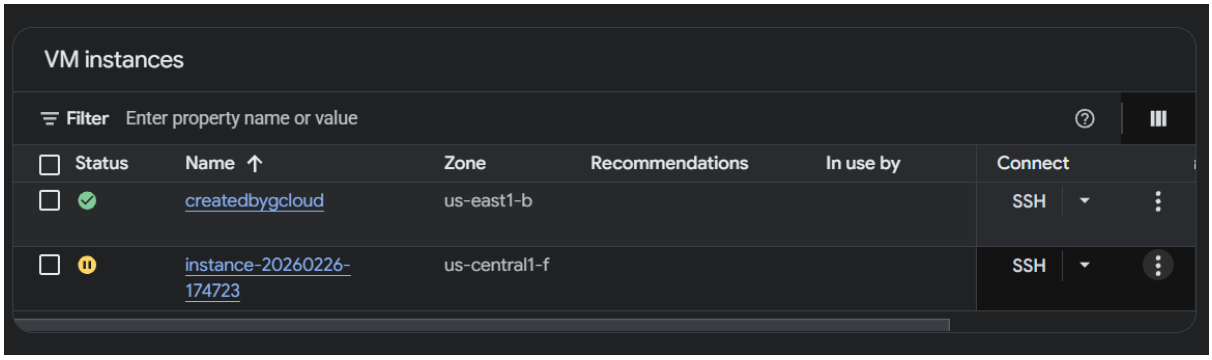
```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud compute instances create createdbygcloud \
    --zone=us-east1-b \0709548-aum-hw1)$ gcloud compute instances create createdbygcloud \
    --machine-type=e2-micro \
    --image-family=debian-12 \
    --image-project=debian-cloud \
    --boot-disk-type=pd-standard \
    --boot-disk-size=10GBtandard \
WARNING: You have selected a disk size of under [200GB]. This may result in poor I/O performance. For more information, see: https://developers.google.com/compute/docs/disks#performance.
Created [https://www.googleapis.com/compute/v1/projects/u0709548-aum-hw1/zones/us-east1-b/instances/createdbygcloud].
WARNING: Some requests generated warnings:
 - You are creating a global DNS VM. VM instances using global DNS are vulnerable to cross-regional outages. To reduce the risk of widespread service disruption, use zonal DNS instead. Learn
more at https://cloud.google.com/compute/docs/networking/zonal-dns

NAME: createdbygcloud
ZONE: us-east1-b
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 10.142.0.2
EXTERNAL_IP: 34.26.108.115
STATUS: RUNNING
```

Question 8) Suspend the VM

Console –

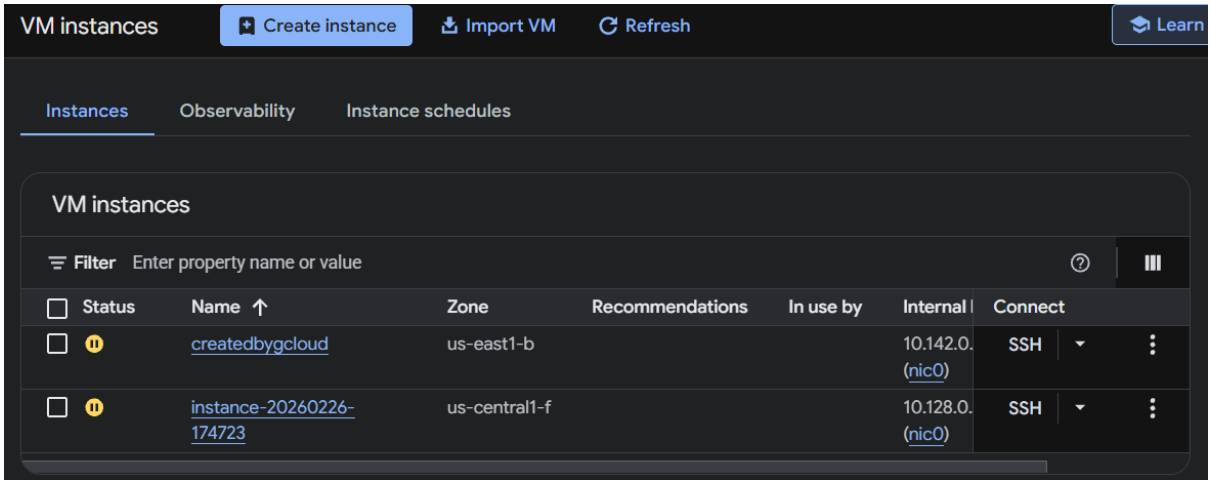






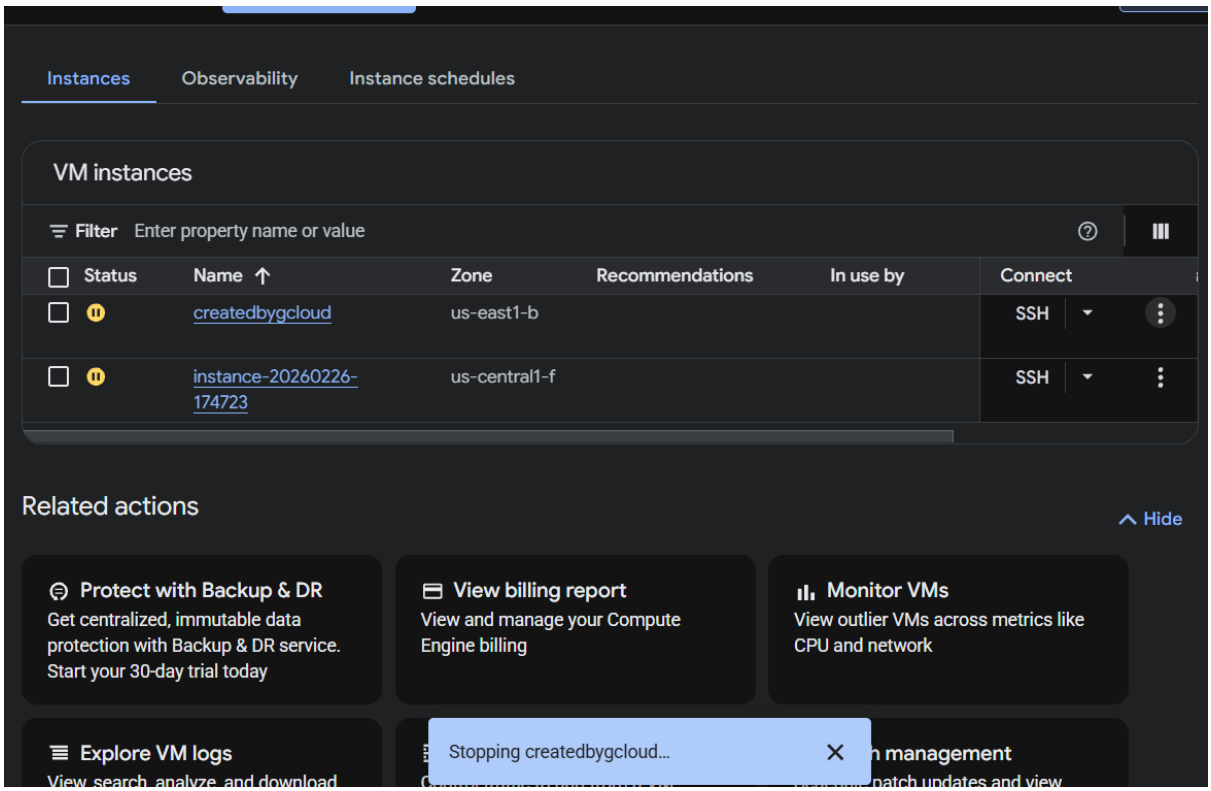
Gcloud –

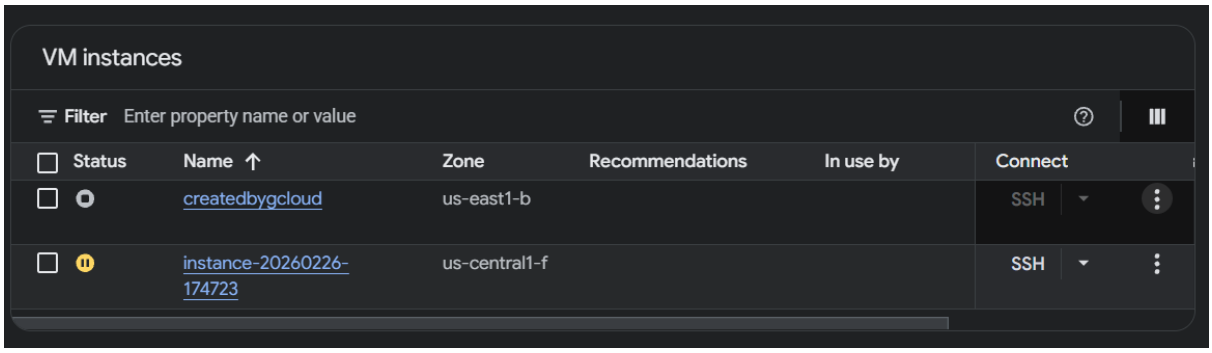
```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud compute instances suspend createdbygcloud --zone=us-east1-b
Suspending instance(s) createdbygcloud...done.                                    edbygcloud --zone=us-east1-b
Updated [https://compute.googleapis.com/compute/v1/projects/u0709548-aum-hw1/zones/us-east1-b/instances/createdbygcloud].
```



Question 9) Stop the VM

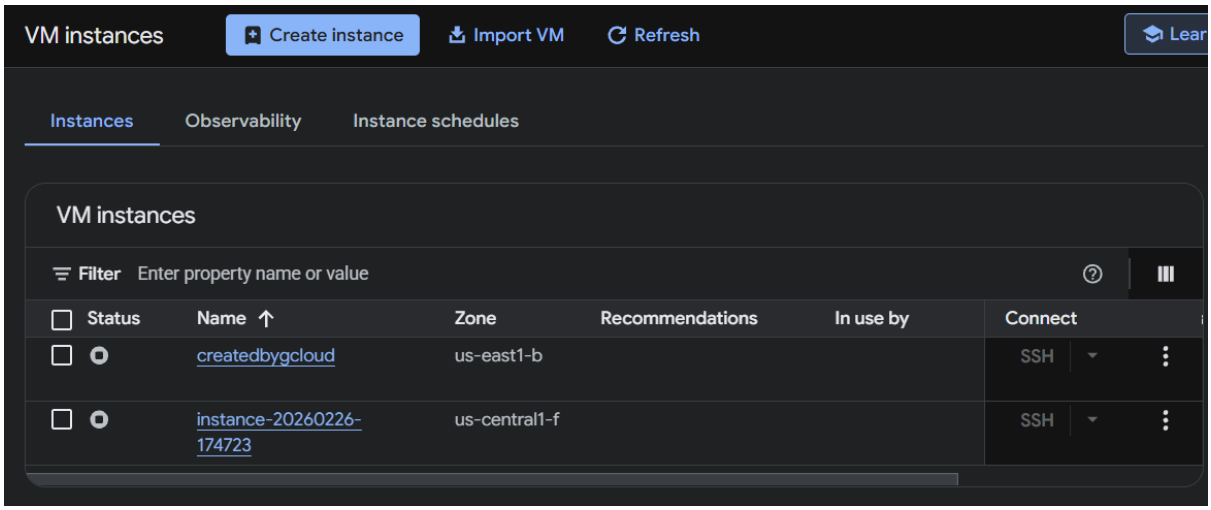
Console –





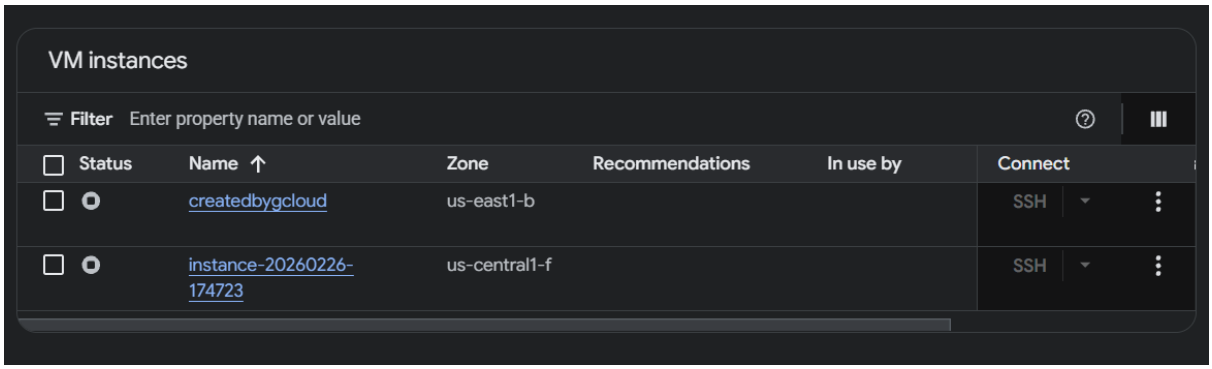
Gcloud –

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud compute instances stop instance-20260226-174723 --zone=us-central1-foud].
Stopping instance(s) instance-20260226-174723...done.                    -20260226-174723 --zone=us-central1-f
Updated [https://compute.googleapis.com/compute/v1/projects/u0709548-aum-hw1/zones/us-central1-f/instances/instance-20260226-174723].
```



Question 10) List all your VM's

Console –

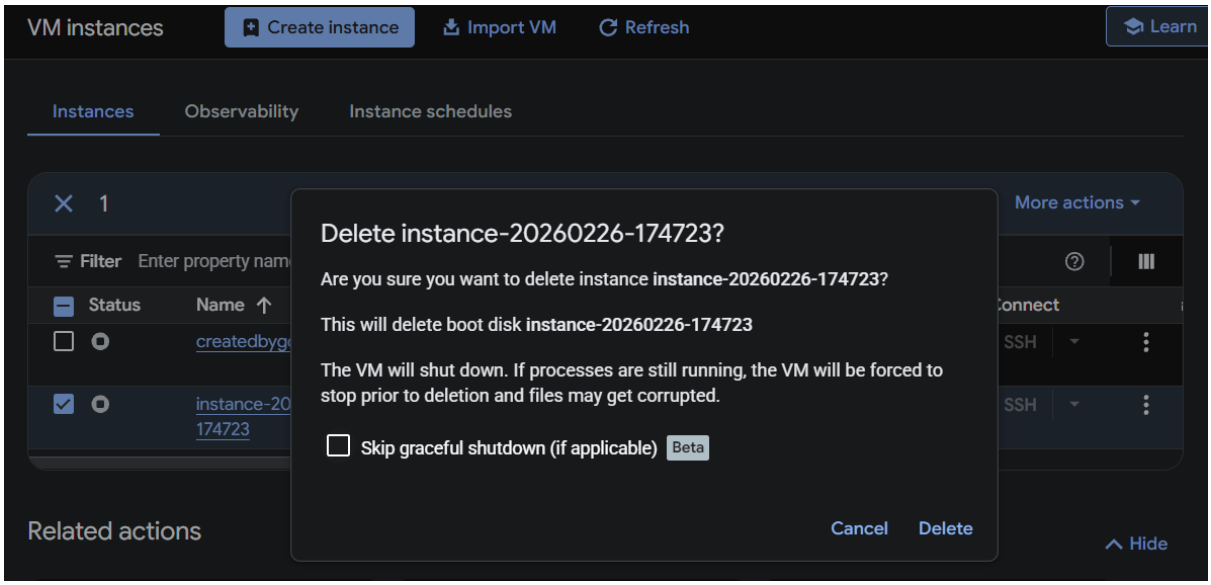


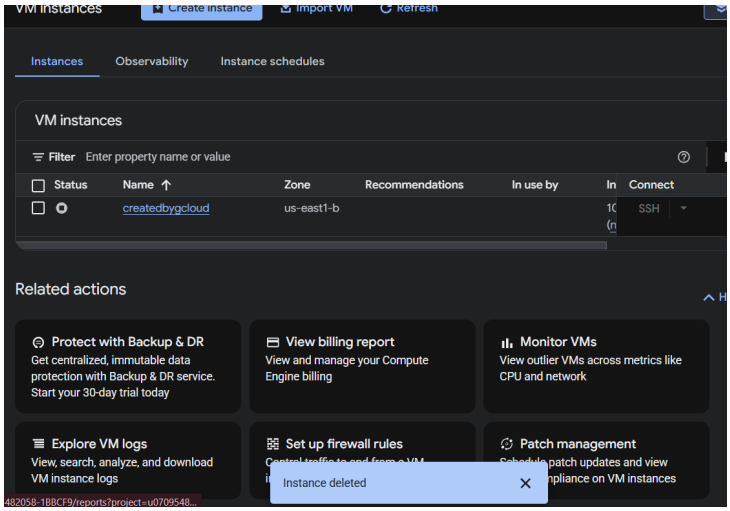
Gcloud –

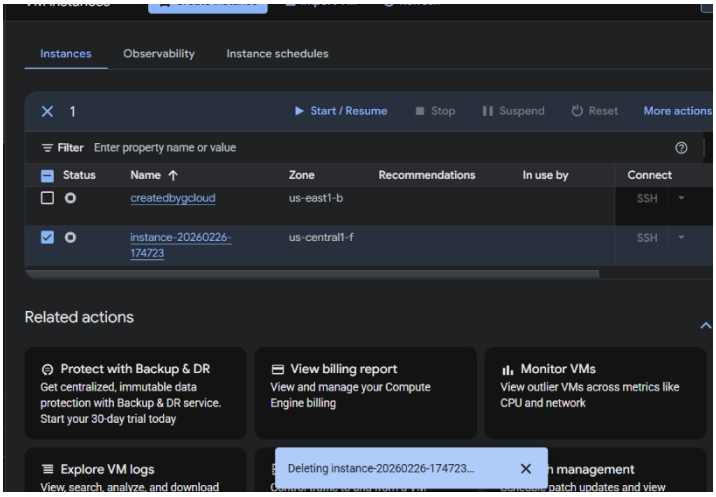
```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud compute instances list
NAME: instance-20260226-174723
ZONE: us-central1-f
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 10.128.0.3
EXTERNAL_IP:
STATUS: TERMINATED

NAME: createdbygcloud
ZONE: us-east1-b
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 10.142.0.2
EXTERNAL_IP:
STATUS: TERMINATED
```

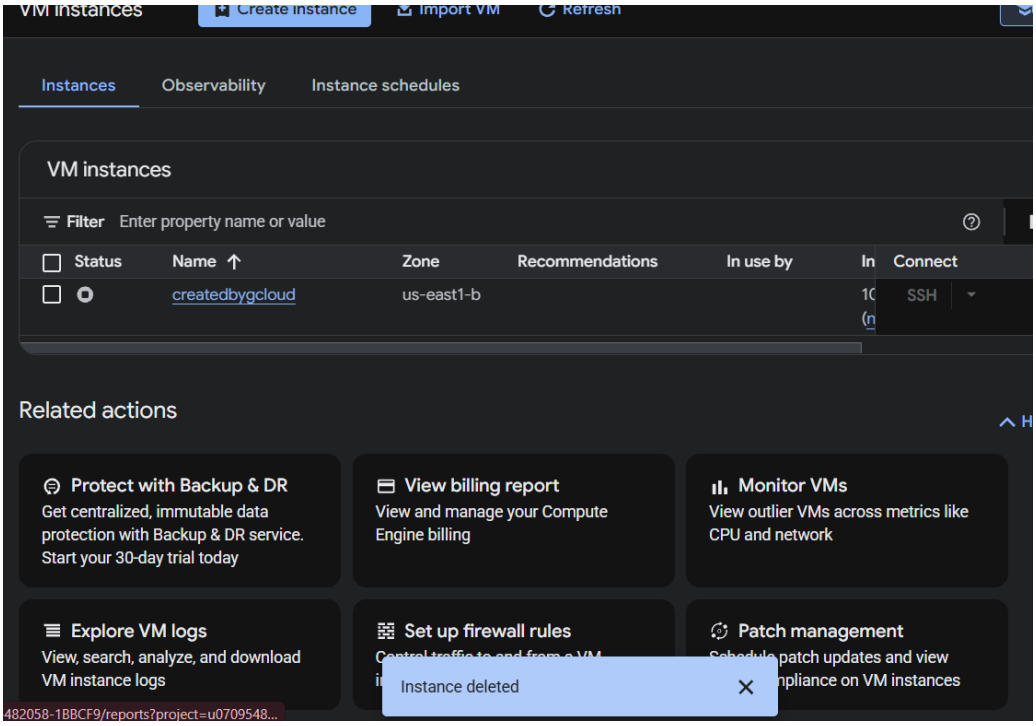
Question 11) Delete the VM

Console –









Gcloud –

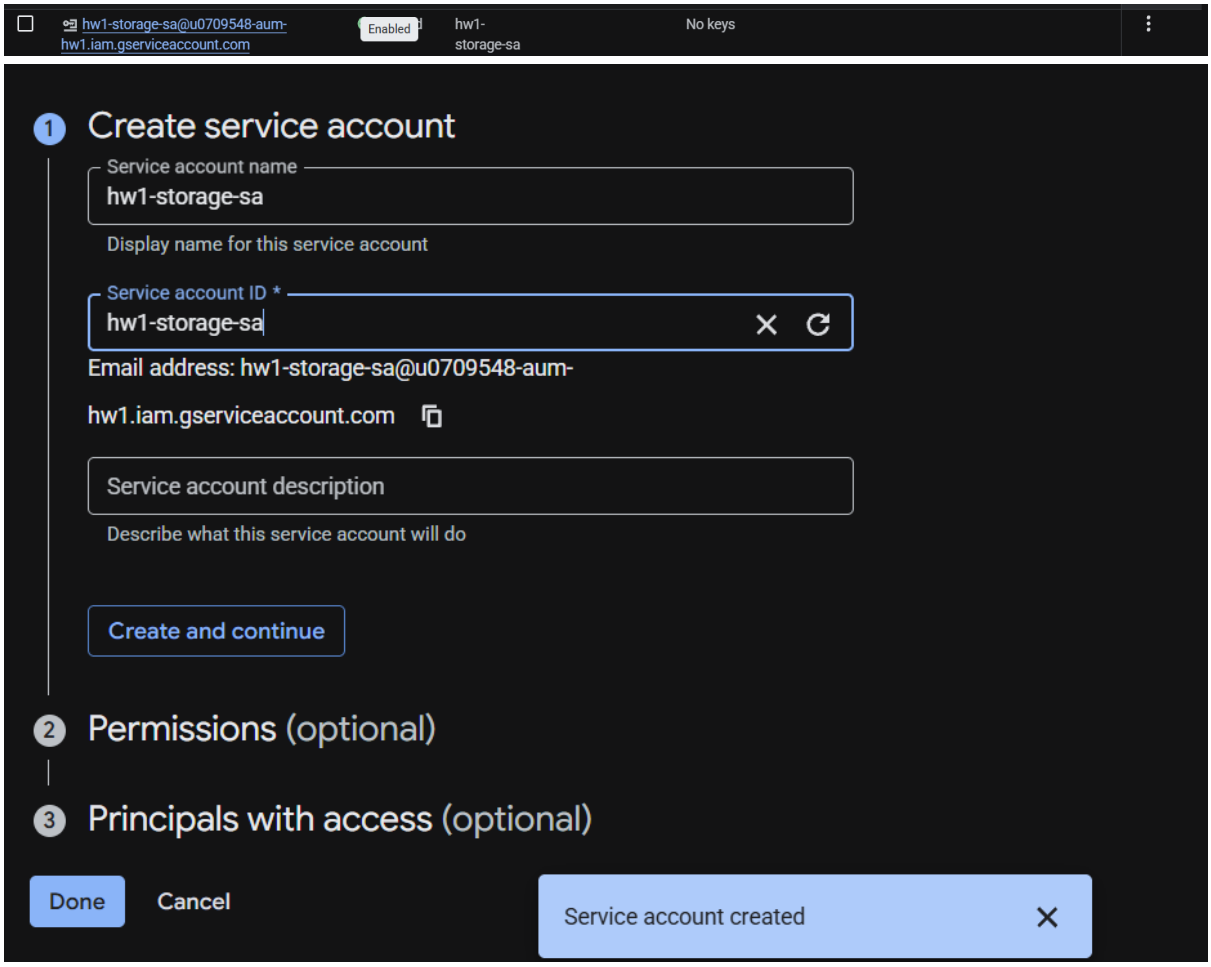
```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud compute instances delete createdbygcloud --zone=us-east1-b
The following instances will be deleted. Any attached disks configured to be tedbygcloud --zone=us-east1-b
auto-deleted will be deleted unless they are attached to any other instances or
the `--keep-disks` flag is given and specifies them for keeping. Deleting a disk
 is irreversible and any data on the disk will be lost.
 - [createdbygcloud] in [us-east1-b]

Do you want to continue (Y/n)?  Y

Deleted [https://www.googleapis.com/compute/v1/projects/u0709548-aum-hw1/zones/us-east1-b/instances/createdbygcloud].
```

```
aum1612@cloudshell:~ (u0709548-aum-hw1)$ gcloud compute instances list
Listed 0 items.
```

All questions below are already done in gcloud (only for console now)

Question 12) Make a service account and give it permissions to access your storage bucket

Console –



Grant access to "aum-hw1-u07079548"

Grant principals access to this resource and add roles to specify what actions the principals can take. Optionally, add conditions to grant access to principals only when a specific criteria is met. Learn more about IAM conditions

Resource

aum-hw1-u07079548

Add principals

Principals are users, groups, domains, or service accounts. Learn more about principals in IAM

New principals *

hw1-storage-sa@u0709548-aum-hw1.iam.gserviceaccount.com

Assign roles

Roles are composed of sets of permissions and determine what the principal can do with this resource. Learn more

Role *

Storage Object Admin

Grants full control over objects, including listing, creating, viewing, and deleting objects.

IAM condition (optional)

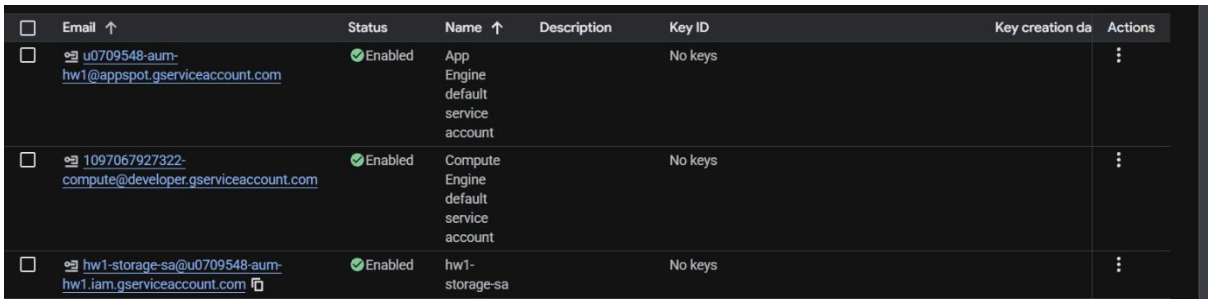
+ Add IAM condition

Save Cancel

	hw1-storage-sa@u0709548-aum-hw1.iam.gserviceaccount.com	hw1-storage-sa	Storage Object Admin

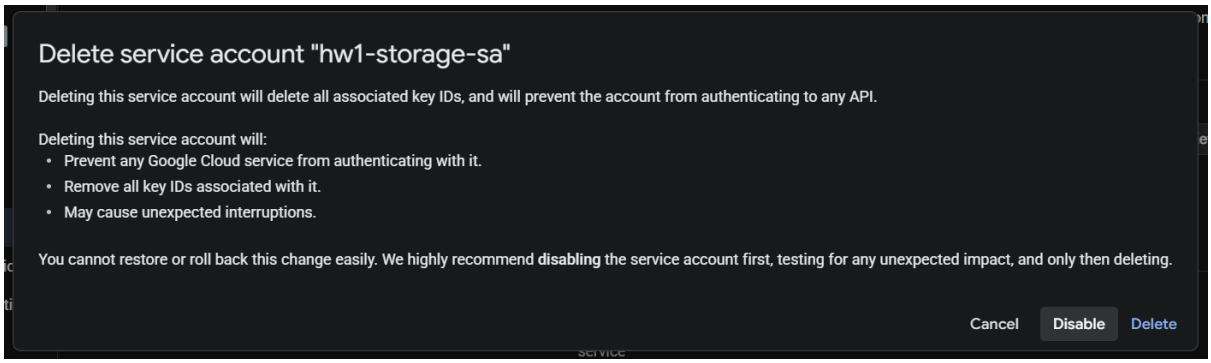
Question 13) List all your service accounts

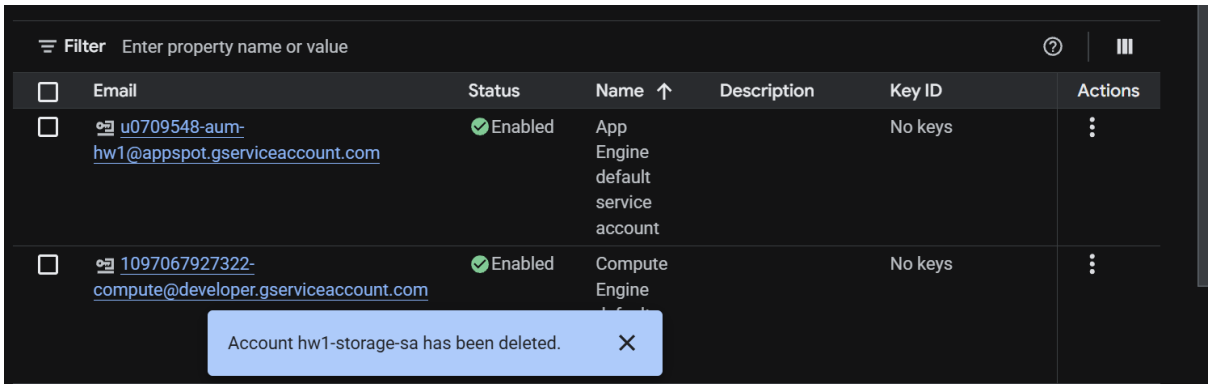
Console –



Email	Status	Name	Description	Key ID	Key creation da	Actions
u0709548-aum-hw1@appspot.gserviceaccount.com	Enabled	App Engine default service account		No keys		
1097067927322-compute@developer.gserviceaccount.com	Enabled	Compute Engine default service account		No keys		
hw1-storage-sa@u0709548-aum-hw1.iam.gserviceaccount.com	Enabled	hw1-storage-sa		No keys		

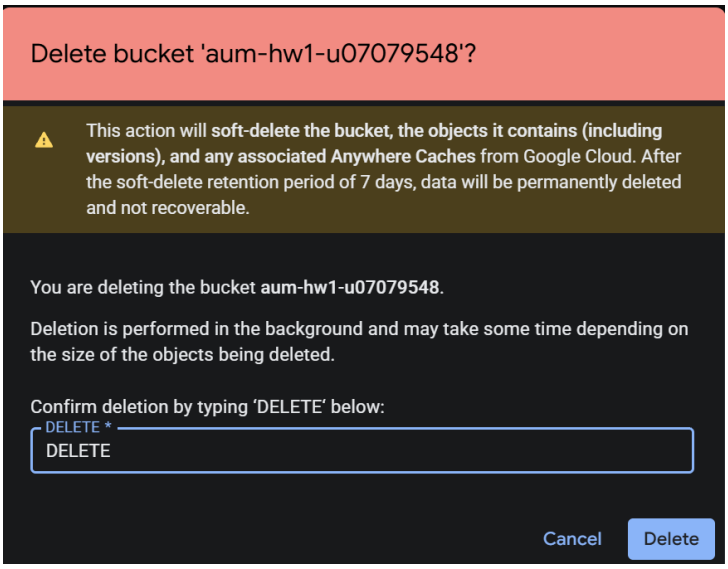
Question 14) Delete the service account





Question 15)

Console –



Filter Filter buckets

Name	Created	Location type
aum-hw2-u07079548	Feb 10, 2026, 12:19:18 AM	Region
gcf-v2-sources-1097067927322-us-c...	Feb 20, 2026, 7:17:21 PM	Region
gcf-v2-uploads-1097067927322.us-c...	Feb 20, 2026, 7:16:26 PM	Region
madewithgcloud	Feb 21, 2026, 12:03:46 AM	Multi-region

Deleted 1 bucket

Gcloud –

```
PS D:\Masters Studies\CS 528 - Cloud Computing\HW\HW1> gcloud storage buckets delete gs://madewithgcloud
|Removing gs://madewithgcloud/...
  Completed 1/1
```

Name	Created	Location type	Location	Default storage class
aum-hw2-u07079548	Feb 10, 2026, 12:19:18 AM	Region	us-central1	Standard
gcf-v2-sources-1097067927322-us-c...	Feb 20, 2026, 7:17:21 PM	Region	us-central1	Standard
gcf-v2-uploads-1097067927322.us-c...	Feb 20, 2026, 7:16:26 PM	Region	us-central1	Standard